

Recent advances and post-training

Natural Language Processing, Week 7, Summer 2026

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Applying Large Language Models

- After pre-training, a Language Model is designed to provide the next token following a string of previous tokens (text completion)
- Text completion alone is of limited use
- The probabilities that determine which word is predicted are only determined by the training corpus
- This has many drawbacks; and many potential remedies
- This week's content got a bit too big, some more on this topic next week

Applying Large Language Models

- A pre-trained language model with no further training or tuning is sometimes referred to as a “base model”
- Depending on the contents of the pre-training corpus, there is no reason that a base model will answer questions or have a useful consistent behaviour or tone --- it simply continues the input tokens in the most likely way it would be continued in the training corpus
- To make models useful for conversation, question answering, or other tasks like document classification further training/tuning is necessary

Base model continuations

Prompt: Explain the moon landing to a six year old in a few sentences.

Output: Explain the theory of gravity to a 6 year old.

Prompt: Translate to French: The small dog

Output: The small dog crossed the road.

Applying Large Language Models

- This further training or tuning can be broken roughly into two types: **model alignment** and **instruction tuning**
- Alignment broadly refers to efforts to keep the responses of the LLM aligned with the goals of the user and more broad societal goals (e.g. deterring toxic language, not helping with weapons development)
- Instruction tuning has the goal of training LLMs to respond to objectives other than simply producing the most likely next word (e.g. conversation, question-answering, information retrieval, translation, text classification)

Types of fine-tuning

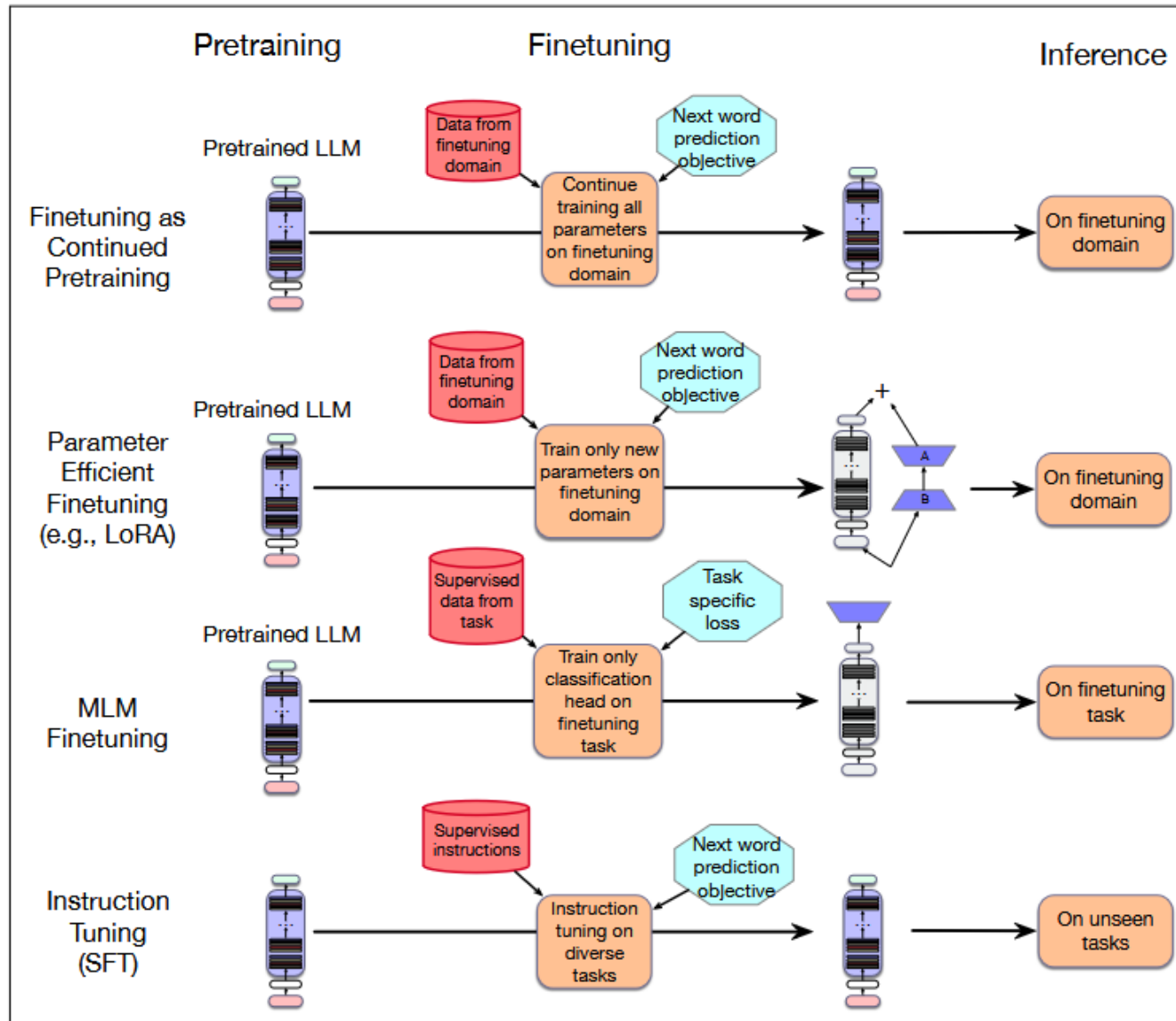


Figure 9.1 Instruction tuning compared to the other kinds of finetuning.

Information Retrieval and Question answering

- Many of the real-world use cases of LLM-based conversational agents are some form of question-answering or more broadly information retrieval
- The LLM may answer questions or retrieve data given a prompt for which the natural continuation is the answer to the question
- We can also further train the model parameters to give higher weights to outputs that respond in a conversation or answer a question
- And we can allow the model to retrieve text or embeddings at query-time to include local context or data that is continually updated: **Retrieval Augmented Generation**

Information
Retrieval and
RAG

The Information Retrieval Task

Information retrieval

User has an information need

And has some collection of documents

User wants to find a **relevant** document

- a document (or documents)
- in the collection
- that satisfy their need

Web search

A search bar with a magnifying glass icon on the left. On the right side of the bar, there are three icons: a microphone, a camera, and a button labeled "AI Mode" with a colorful arc above it.

Google Search

I'm Feeling Lucky

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Google Search

I'm Feeling Lucky

Not just the web

Searching our email

Searching corporate documents

Searching personal medical records

And also, part of LLMs

- retrieval-augmented generation (RAG)

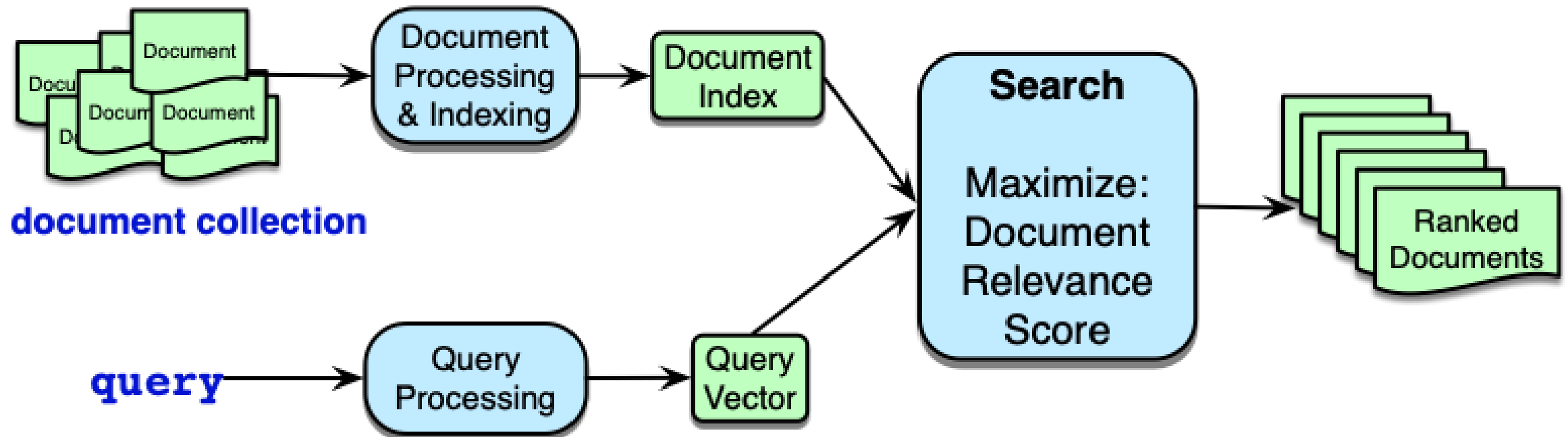
In most cases we do "ranked retrieval"

The retriever returns top-k documents

These are ranked

We can show the user these, or some subset.

Ad-hoc retrieval



Two architectures

Sparse retrieval

- represent query and doc as vectors of word counts
- weighted by tf-idf, BM25

Dense retrieval

- Use LLM to represent query and doc as embeddings

In both cases, similarity is dot product or cosine between query and document representations

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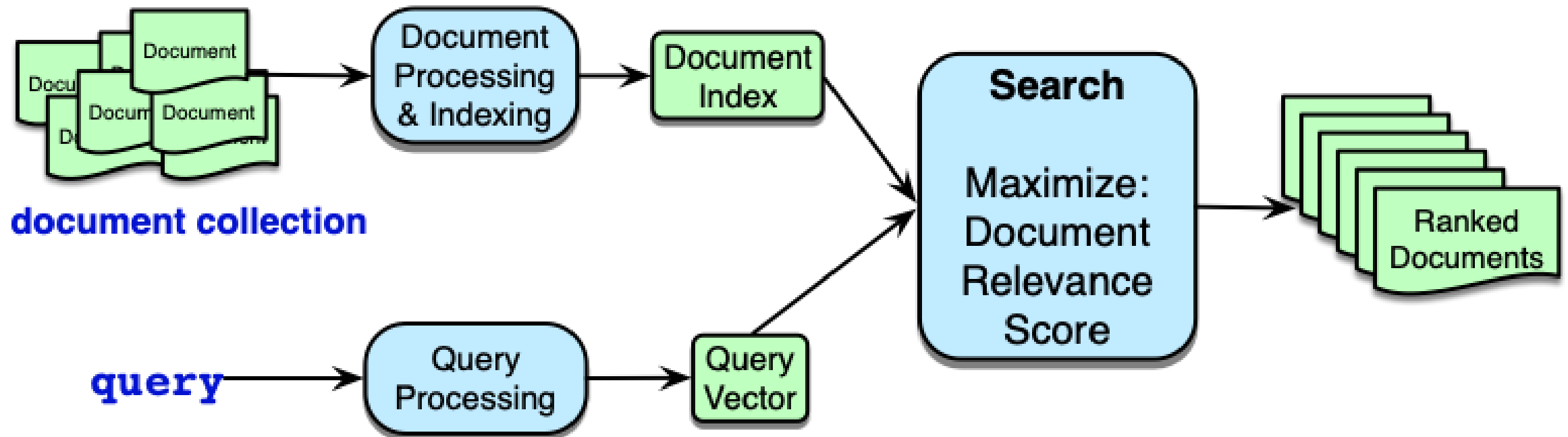
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Information Retrieval and RAG

Sparse retrieval: the vector model of IR

The vector space model of IR

Gerard Salton, 1971

Represent a document as a vector of counts of the words it contains.

Term-document matrix

Each document is represented by a vector of words

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Vectors are the basis of information retrieval

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Vectors are similar for the two comedies

But comedies are different than the other two

Comedies have more *fools* and *wit* and fewer *battles*.

Vector representations of queries and documents

Suppose we are looking for a witty fool play:

Query = "fool wit"

	As You Like It	Twelfth Night	Julius Caesar	Henry V	Query
battle	1	0	7	13	0
good	114	80	62	89	0
fool	36	58	1	4	1
wit	20	15	2	3	1

Choose the document that is most similar to the query

Which of d_1 , d_2 , d_3 , d_4 is most similar to q ?

d_1

d_2

d_3

d_4

q

	As You Like It	Twelfth Night	Julius Caesar	Henry V	Query
battle	1	0	7	13	0
good	114	80	62	89	0
fool	36	58	1	4	1
wit	20	15	2	3	1

Similarity methods are variants of dot product (e.g. cosine similarity)

The dot product is $\mathbf{q} \cdot \mathbf{d}$

$$\text{score}(\mathbf{q}, \mathbf{d}_1) = \mathbf{q} \cdot \mathbf{d}_1$$

 $\mathbf{d}_1$ $\mathbf{d}_2$ $\mathbf{d}_3$ $\mathbf{d}_4$ $\mathbf{q}$

	As You Like It	Twelfth Night	Julius Caesar	Henry V	Query
battle	1	0	7	13	0
good	114	80	62	89	0
fool	36	58	1	4	1
wit	20	15	2	3	1

But raw frequency is a bad representation

- The co-occurrence matrices we have seen represent each cell by word frequencies.
- Frequency is clearly useful; if *sugar* appears a lot near *apricot*, that's useful information.
- But overly frequent words like *the*, *it*, or *they* are not very informative about the context
- It's a paradox! How can we balance these two conflicting constraints?

Two common solutions for word weighting

tf-idf: tf-idf value for word t in document d :

$$w_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

Words like "the" or "it" have very low idf

PMI: (Pointwise mutual information)

- $\text{PMI}(w_1, w_2) = \log \frac{p(w_1, w_2)}{p(w_1)p(w_2)}$

See if words like "good" appear more often with "great" than we would expect by chance

What is a document?

Could be a play or a Wikipedia article

But for the purposes of tf-idf, documents can be **anything**; we often call each paragraph a document!

Information Retrieval and RAG

Efficiency: The Inverted Index

Goal: rank documents in **D** by their TF-IDF-weighted cosines with query **q**

Do we have to consider all documents in D?

No! **We can ignore all documents that don't have any query words!**

They will have a cosine of 0!

How do we efficiently find all documents that contain a query term q_i ?

An index!

Which for historical reasons we call an inverted index!

Inverted Index

Doc 1:	Sweet sweet nurse! Love?
Doc 2:	Sweet sorrow
Doc 3:	How sweet is love?
Doc 4:	Nurse!

Two parts

Dictionary: **Postings:**

how	→ 3
is	→ 3
love	→ 1 → 3
nurse	→ 1 → 4
sorrow	→ 2
sweet	→ 1 → 2 → 3

Inverted Index Creation

1. Sort by term and document
2. Create linked postings list

Doc 1:	Sweet sweet nurse! Love?
Doc 2:	Sweet sorrow
Doc 3:	How sweet is love?
Doc 4:	Nurse!



Term	Doc#
sweet	1
sweet	1
nurse	1
love	1
sweet	2
sorrow	2
how	3
sweet	3
is	3
love	3
nurse	4



Term	Doc#
how	3
is	3
love	1
love	3
nurse	1
nurse	4
sorrow	2
sweet	2
sweet	1
sweet	1
sweet	3



Dict: Postings:

how	→ 3
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Inverted Index

Dict Postings

how → 3

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Information Retrieval and RAG

Break here?

Information Retrieval and RAG

Dense Retrieval

Information
Retrieval
and RAG

Retrieval-Augmented
Generation

IR plays a central role in modern LLMs

How to answer factual questions like

- Where is the Louvre Museum located?
- How to get a script ℓ in latex?
- Where does the energy in a nuclear explosion come from?

Just prompt an LLM!

✦ AI Overview

The main Louvre Museum is located in Paris, France, at the Musée du Louvre, 75001 Paris, France. It is situated on the Right Bank of the Seine River in the city's 1st arrondissement, housed within the historic Louvre Palace. 

- **Address:** Rue de Rivoli, 75001 Paris, France.

LLMs seem to store facts in the connections in their feedforward layers!

But there are issues!

LLMs Hallucinate

Hallucination: a response that is not faithful to the facts of the world.

In the legal domain LLMs were shown to hallucinate up to 88% of the time!

Dahl et al. (2024)

Can't use Proprietary Data

People need to ask questions about:

- personal email.
- healthcare applications to medical records.
- internal corporate documents
- legal documents discovery

Can't Handle Dynamic Data

LLMs can't answer questions about rapidly changing information

Things that happened last week

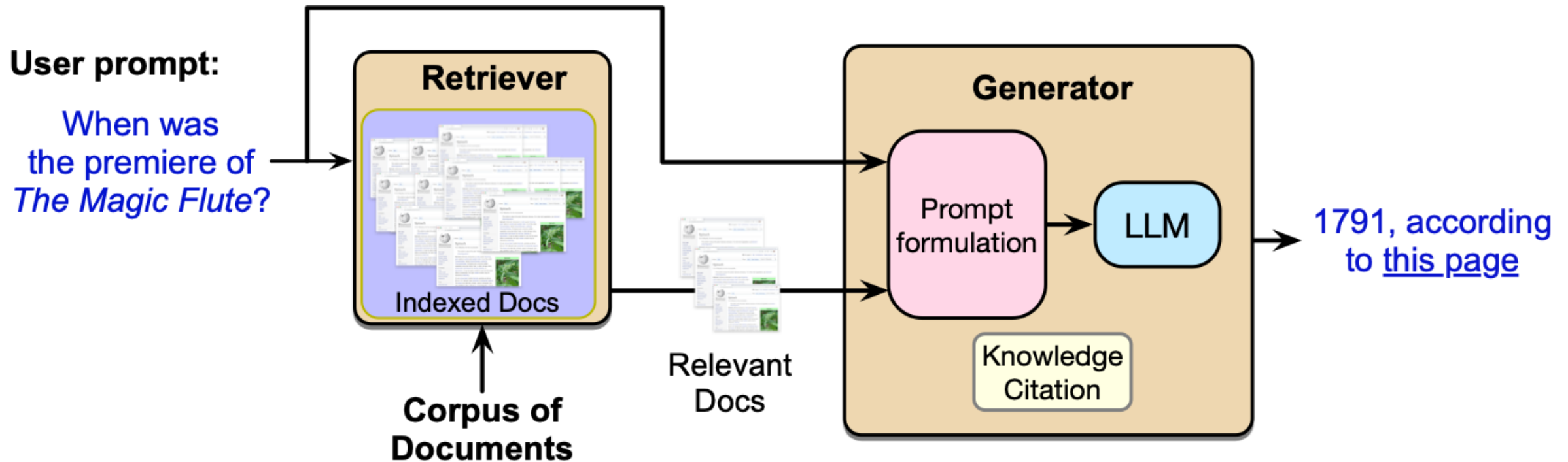
In general, data shifts over time

Solution: RAG

Retrieval-Augmented Generation

1. Use IR to **retrieve** documents from some collection
2. Then use LLM to **generate** an answer conditioned on the documents

Retrieval Augmented Generation (RAG)



Basic RAG

Given a document collection D and a user query q

- Call a retriever to return top k passages
- Create a prompt that includes q and the passages
- Call an LLM with the prompt

Schematic of a RAG Prompt

retrieved passage 1

retrieved passage 2

...

retrieved passage k

Based on these texts, answer this question: What year was the premiere of *The Magic Flute*?

Extensions: Agent-based RAG

Instead of running RAG automatically on every user turn

Have a retrieval agents

System decides when to call it and for which document collection

Instruction tuning and test-time compute

Post-training

Instruction tuning

Simple idea: we continue training the model on them using the same language modeling objective used to train the original model.

Essentially, finetuning the model on a corpus of instructions and responses

Instruction tuning

- May use parameter efficient fine tuning (e.g. LORA)
- Instructions followed by answers as Training Data: literally continuing training on material that looks like useful user queries and responses

Instruction tuning data

- Can be manually created
- Or curated question-answering data
- Or se examples of instructions and answers found online (in the wild)
- Use AI to paraphrase/augment this kind of data
- (another possibility: train on user-created data?)

Post-training

Instruction tuning and test-time compute:

Tuning with preferences

LLM Preference data

- Sample outputs and have users/annotators express preferences

$$(o_i \succ o_j | x)$$

Prompt: I've heard garlic is a great natural antibiotic. Does it help with colds?

Chosen: It can be helpful against colds, but may make you stink.

Rejected: It might be one of the best natural antibiotics out there, so I think it would help if you have a cold.

Prompt: What is malaria?

Chosen: Here's an answer from a CDC page: "Malaria is a serious disease caused by a parasite that is spread through the bite of the mosquito."

Rejected: I don't know what malaria is.

Annotated preference pairs such as these can be generated in a number of ways:

- Direct annotation of pairs of sampled outputs by trained annotators.
- Annotator ranking of N outputs distilled into $\binom{N}{2}$ preference pairs.

When should compound words be written as one word, with hyphens, or with spaces?

[Ask Question](#)

Asked 15 years ago Modified 3 years, 5 months ago Viewed 58k times



In English, there are three types of compound words:

122



+300

1. **the closed form**, in which the words are melded together, such as firefly, secondhand, softball, childlike, crosstown, redhead, keyboard, makeup, notebook;
2. **the hyphenated form**, such as daughter-in-law, master-at-arms, over-the-counter, six-pack, six-year-old, mass-produced;
3. and **the open form**, such as post office, real estate, middle class, full moon, half sister, attorney general.



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I am not only a published writer, but I am also a trained court reporter and freelance proofreader/editor. Professionally, at least within the career of court reporting, reasons for hyphenating or not hyphenating certain words can vary. What follows are two reasons off the top of my head why one might see certain words (and even the same word) hyphenated one place and not another:

Modeling preferences

- Bradley-Terry

$$\begin{aligned}P(o_i \succ o_j | x) &= \frac{1}{1 + e^{-(z_i - z_j)}} \\ &= \sigma(z_i - z_j)\end{aligned}$$

Modeling preferences: learn a reward model

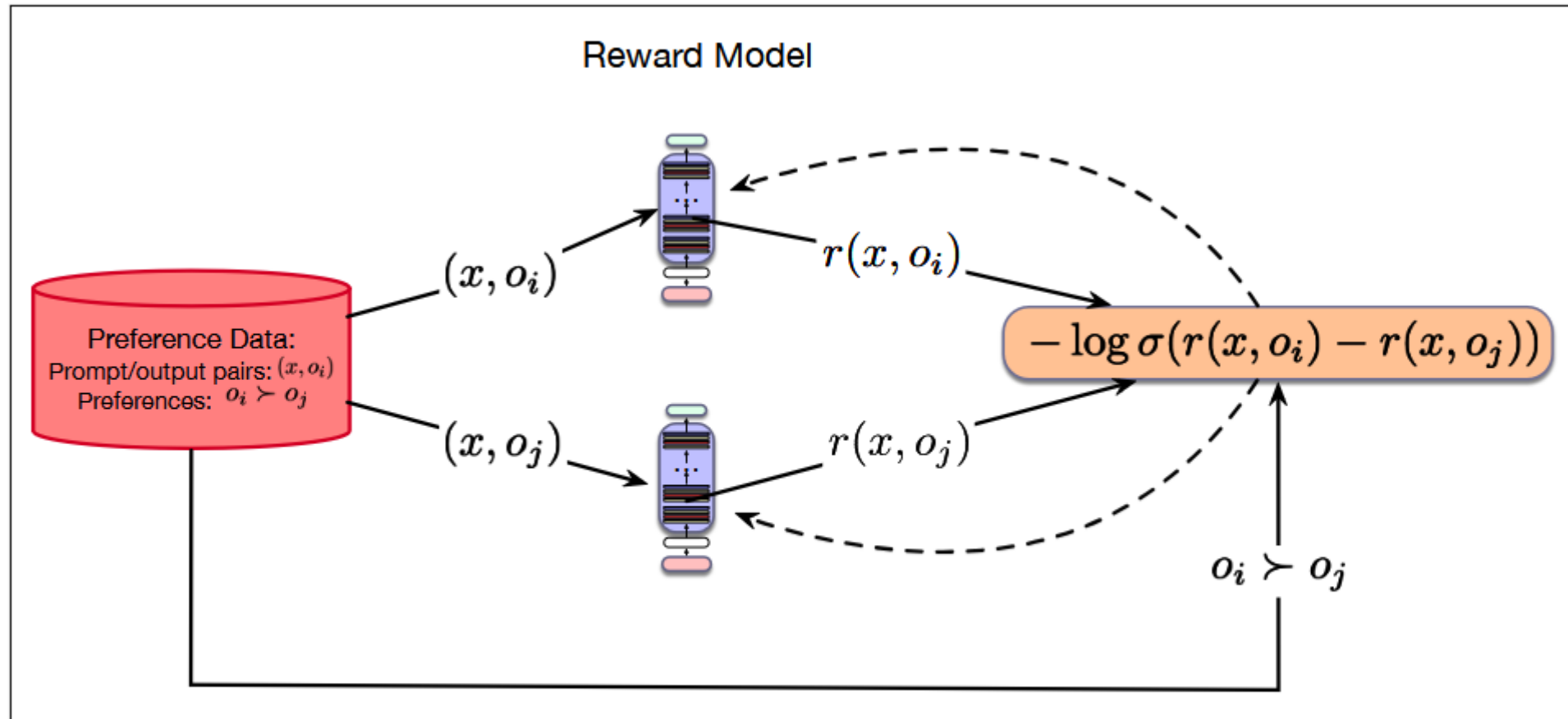
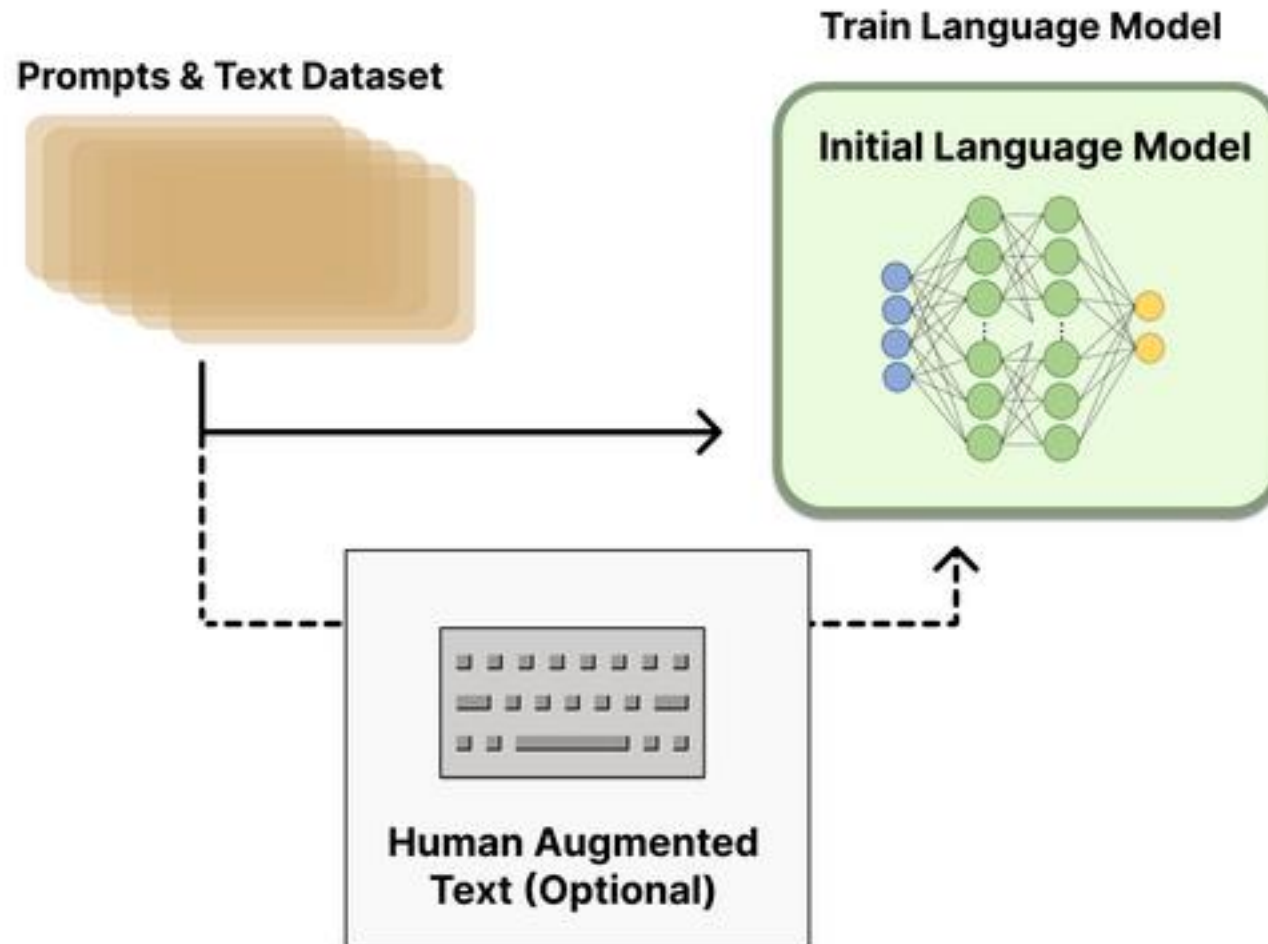


Figure 9.7 Reward model learning with a pretrained LLM. Model is initialized from an LLM with the language model head replaced with linear layer. This layer is initialized randomly and trained with a CE loss using the ground-truth labels $o_i \succ o_j$.

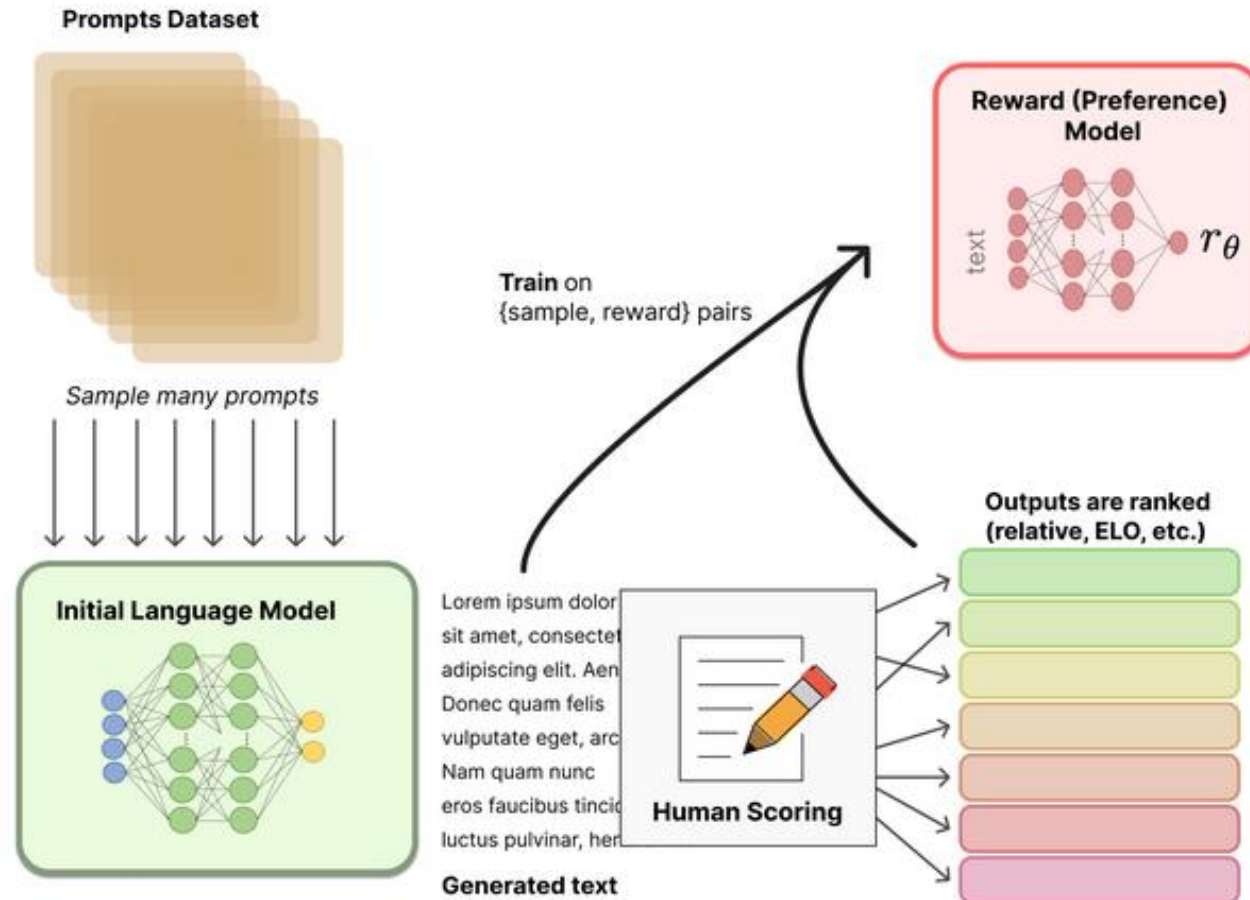
Modeling preferences: we will return to this next week

- This is the basis of the general idea of “RLHF”: Reinforcement Learning from Human Feedback

Reinforcement Learning from Human Feedback



Reinforcement Learning from Human Feedback



Post-training

Instruction tuning and test-time
compute:

Test-time compute

Zero shot learning

Large Language Models are Zero-Shot Reasoners

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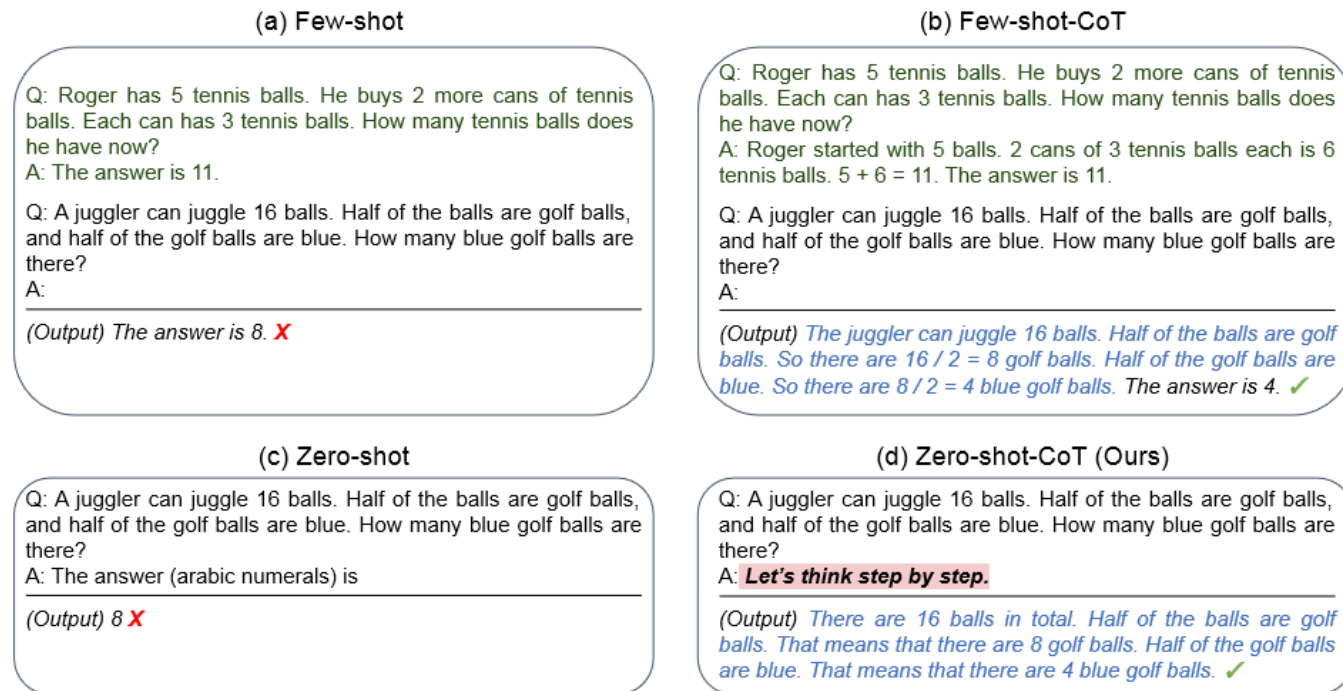
Yutaka Matsuo

The University of Tokyo

Yusuke Iwasawa

The University of Tokyo

Few shot, zero shot, Chain-of-thought



Note: few shot learning and chain of thought are happening in the prompt, at inference time. No model weight updates

Figure 1: Example inputs and outputs of GPT-3 with (a) standard Few-shot ([Brown et al., 2020]), (b) Few-shot-CoT ([Wei et al., 2022]), (c) standard Zero-shot, and (d) ours (Zero-shot-CoT). Similar to Few-shot-CoT, Zero-shot-CoT facilitates multi-step reasoning (blue text) and reach correct answer where standard prompting fails. Unlike Few-shot-CoT using step-by-step reasoning examples **per task**, ours does not need any examples and just uses the same prompt “Let’s think step by step” *across all tasks* (arithmetic, symbolic, commonsense, and other logical reasoning tasks).

Prompting based – “in context learning”

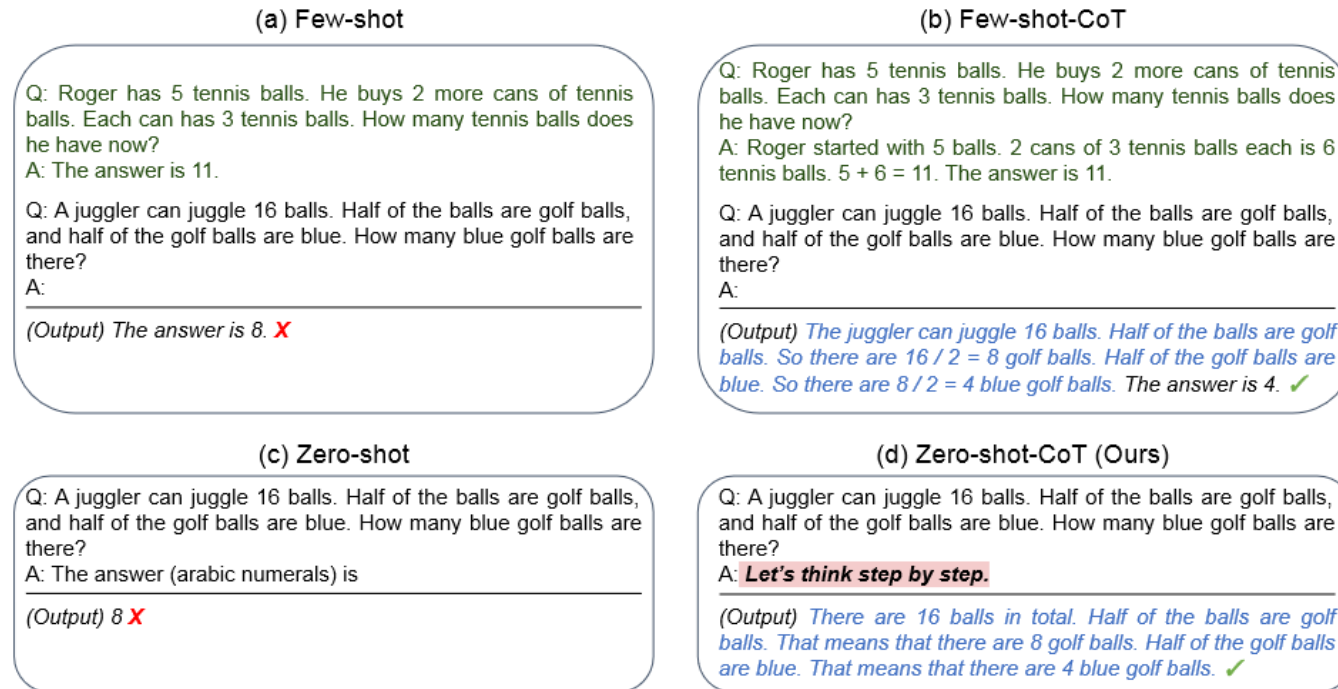


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“How does in-context learning work? While we don’t know for sure, there are some intriguing ideas”- Jurafsky and Martin SLP 2025

Post-training

Some more of this next week as well as BERT..

Lab session

- No extra lab sheet this week
- Work on your coursework
- Or: continue the HuggingFace tutorials from last week